

# MCU – ESP32-S3-WROOM-1 N16R8

Dual-core LX7 @ 240MHz, 16MB Flash, 8MB PSRAM

## GPIO ASSIGNMENT TABLE

**Display (8080 parallel):**  
GPIO4-11 = D0-D7 (8-bit data bus)  
GPIO12=CS GPIO13=RST GPIO14=DC  
GPIO46=WR (RD/BL tied +3V3 on FPC)

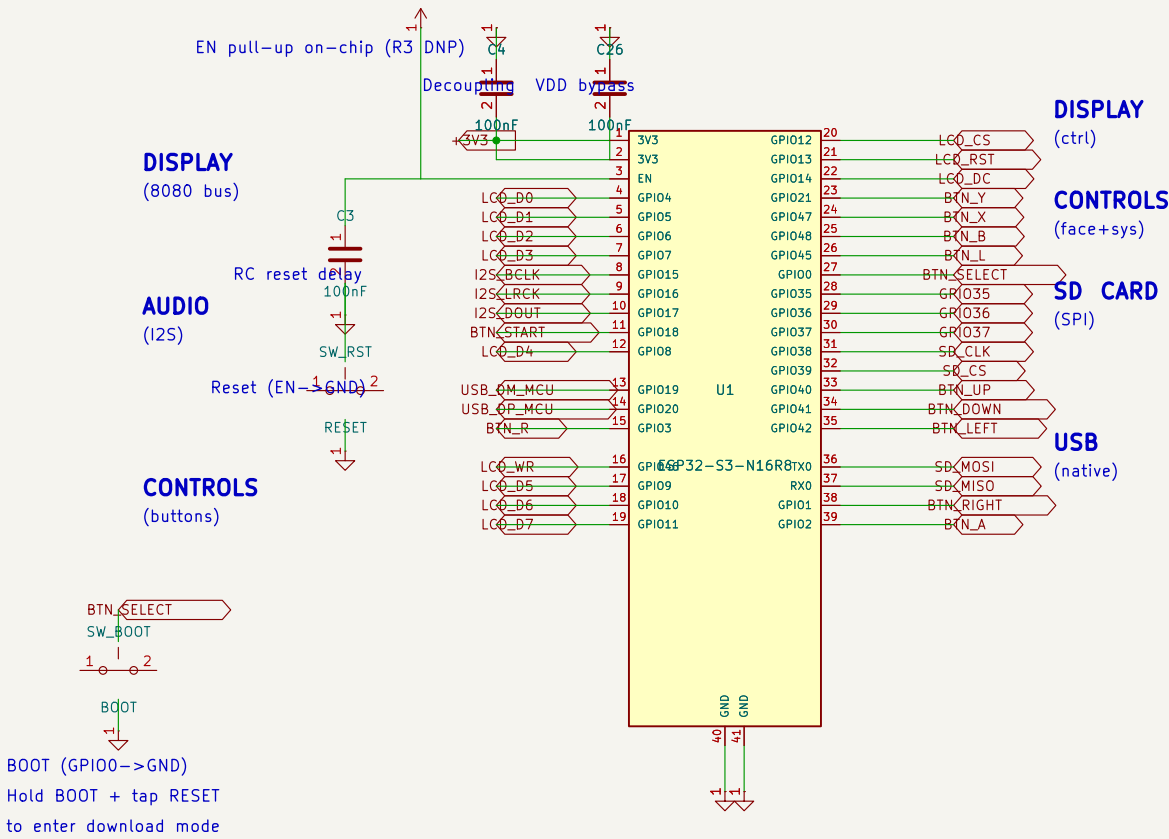
**Audio (I2S):**  
GPIO15=BCLK GPIO16=LRCK GPIO17=DOUT

**SD Card (SPI):**  
GPIO44=MOSI GPIO43=MISO GPIO38=CLK GPIO39=CS

**Controls (active low, 10k pull-up):**  
D-pad: GPIO40=UP 41=DOWN 42=LEFT 1=RIGHT  
Face: GPIO2=A 48=B 47=X 21=Y  
Sys: GPIO18=START 0=SELECT 45=L 3=R

**USB (native, firmware flash + debug):**  
GPIO19=D- GPIO20=D+

**Reserved (do not use, module-internal):**  
GPIO26-32 = Octal Flash (N16)  
GPIO33-37 = Octal PSRAM (R8)



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ESP32 Emu Turbo – Handheld Retro Console

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File: 02-mcu.kicad\_sch

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